



Calculus of Several Variables

MATH 252

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Course Outline

- Vectors and Geometry of Space
- Functions of Several Variables
- Differentiation and Optimization
- Multiple Integration
- Vector Calculus

Reading book: Calculus III by Paul Dawkins



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Lecture One



Vectors and Geometry of Space



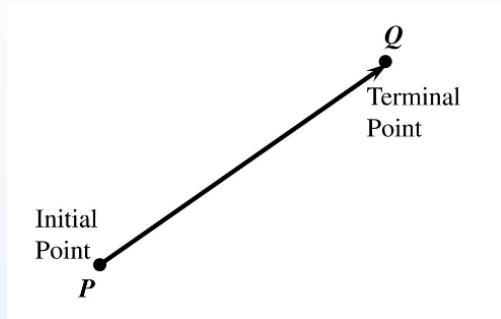
Definition

- A **vector** is any quantity that has both **magnitude** and **direction**.
- Some examples of vectors include: acceleration, force, velocity, displacement, temperature, magnetic polarization, etc.
- Vectors can be used by air-traffic controllers when tracking planes, by meteorologists when describing wind conditions.
- Computer programmers use vectors in designing virtual worlds.



Vectors In The Plane

- Given the line segment as shown in the figure:



- The **magnitude** indicates the **distance covered** by the object.
- The **direction** indicates the **initial** point to the **terminal** point.



Cont.

- Consider the figure below

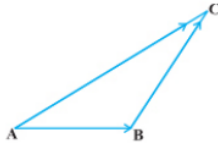


Fig. 1

- The displacement of the object from A to B , followed by displacing it from B to C , is the same as displacing it from A to C .
- We call this the **resultant vector**:

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}.$$

Operations and Properties on Vectors

Given the vectors $\vec{a} = \langle a_1, a_2 \rangle$, $\vec{b} = \langle b_1, b_2 \rangle$, and a scalar c , the following properties hold:

- $\|\vec{a}\| = \sqrt{a_1^2 + a_2^2}$ (magnitude).
- $\langle a_1, a_2 \rangle + \langle b_1, b_2 \rangle = \langle a_1 + b_1, a_2 + b_2 \rangle$.
- $\langle a_1, a_2 \rangle - \langle b_1, b_2 \rangle = \langle a_1 - b_1, a_2 - b_2 \rangle$.



$$\begin{aligned}c \|\vec{a}\| &= \|\langle ca_1, ca_2 \rangle\| \\&= \sqrt{(ca_1)^2 + (ca_2)^2} \\&= \sqrt{c^2 a_1^2 + c^2 a_2^2} \\&= |c| \sqrt{a_1^2 + a_2^2} \\&= |c| \|\vec{a}\|\end{aligned}$$



Vector Arithmetic

Given the vectors,

$$\vec{a} = \langle 2, 3 \rangle \quad \text{and} \quad \vec{b} = \langle 1, 4 \rangle,$$

find the following:

1. $\vec{a} + \vec{b}$
2. $-5\vec{a}$
3. $3\vec{a} - 2\vec{b}$
4. $\|3\vec{a} - 2\vec{b}\|$



Solution

1.

$$\vec{a} + \vec{b} = \langle 2, 3 \rangle + \langle 1, 4 \rangle = \langle 2 + 1, 3 + 4 \rangle = \langle 3, 7 \rangle$$

2.

$$-5\vec{a} = -5\langle 2, 3 \rangle = \langle -5(2), -5(3) \rangle = \langle -10, -15 \rangle$$

3.

$$3\vec{a} - 2\vec{b} = 3\langle 2, 3 \rangle - 2\langle 1, 4 \rangle = \langle 6, 9 \rangle - \langle 2, 8 \rangle = \langle 4, 1 \rangle$$

4.

$$\|3\vec{a} - 2\vec{b}\| = \|\langle 4, -1 \rangle\| = \sqrt{(4)^2 + (-1)^2} = \sqrt{17}$$



Exercise

Given the vectors $\vec{a} = \langle 7, -1 \rangle$ and $\vec{b} = \langle 6, 5 \rangle$,

compute:

- $2\vec{a} + 5\vec{b}$,
- $-3\vec{a}$,
- $3\vec{a} + 5\vec{b}$,
- $2\vec{a} - 7\vec{b}$,
- $\|5\vec{a} - 3\vec{b}\|$.



Zero Vector

- A zero vector is a vector that has all its elements as zero.
- A two-dimensional (2D) zero vector is given as:

$$\langle 0, 0 \rangle = \vec{0}$$

Properties of Zero Vectors:

- Has a length of 0.
- Has no direction.



Parallel Vectors

- Two vectors having the same or opposite direction are called parallel vectors.
- Two vectors $\vec{a}$ and $\vec{b}$ are said to be parallel if there exists $k \in \mathbb{R}$ such that:

$$\vec{a} = k \vec{b}.$$

- The zero vector is parallel to every vector.



Two non-parallel
Vectors



Parallel Vectors in
same direction



Parallel Vectors in
opposite direction



Example

Determine whether the given vectors are parallel or not:

1. $\vec{p} = \langle 2, 5 \rangle$ and $\vec{q} = \langle 10, 25 \rangle$,

2. $\vec{p} = \langle 3, 5 \rangle$ and $\vec{q} = \langle 7, 15 \rangle$.



Solution

1. For $\vec{p} = \langle 2, 5 \rangle$ and $\vec{q} = \langle 10, 25 \rangle$:

$$\vec{q} = c\vec{p}, \quad \langle 10, 25 \rangle = c\langle 2, 5 \rangle.$$

$$\langle 10, 25 \rangle = \langle 2c, 5c \rangle.$$

Thus:

$$10 = 2c \implies c = 5, \quad 25 = 5c \implies c = 5.$$

Therefore, the two vectors are **parallel**.

2. For $\vec{p} = \langle 3, 5 \rangle$ and $\vec{q} = \langle 7, 15 \rangle$:

$$\vec{q} = c\vec{p}, \quad \langle 7, 15 \rangle = c\langle 3, 5 \rangle.$$

$$\langle 7, 15 \rangle = \langle 3c, 5c \rangle.$$

Thus:

$$7 = 3c \implies c = \frac{7}{3}, \quad 15 = 5c \implies c = 3.$$

Since c cannot have multiple values at the same time, the vectors are **not parallel**.



Properties of Vector Arithmetic

Given any two-dimensional vector space:

$$V_2 = \{\langle x, y \rangle \mid x, y \in \mathbb{R}\},$$

the following properties hold for all $\vec{a}, \vec{b}, \vec{c} \in \mathbb{R}^2$ and for any scalars $m, e \in \mathbb{R}$:

- $\vec{a} + \vec{b} = \vec{b} + \vec{a}$ (commutativity),
- $\vec{a} + (\vec{b} + \vec{c}) = (\vec{a} + \vec{b}) + \vec{c}$ (associativity),
- $\vec{a} + 0 = \vec{a}$ (zero vector),
- $\vec{a} + (-\vec{a}) = 0$ (additive inverse).



- $m(\vec{a} + \vec{b}) = m\vec{a} + m\vec{b}$ (distributive property),
- $e \cdot \vec{a} = \vec{a}$ (multiplicative identity),
- $e + \vec{a} = \vec{a}$ (additive identity).



Example

Given $\vec{a} = \langle a_1, a_2 \rangle$ and $\vec{b} = \langle b_1, b_2 \rangle$,

prove that:

$$\vec{a} + \vec{b} = \vec{b} + \vec{a}.$$



Solution

Given $\vec{a} = \langle a_1, a_2 \rangle$ and $\vec{b} = \langle b_1, b_2 \rangle$:

- To prove $\vec{a} + \vec{b}$:

$$\vec{a} + \vec{b} = \langle a_1, a_2 \rangle + \langle b_1, b_2 \rangle = \langle a_1 + b_1, a_2 + b_2 \rangle.$$

- To prove $\vec{b} + \vec{a}$:

$$\vec{b} + \vec{a} = \langle b_1, b_2 \rangle + \langle a_1, a_2 \rangle = \langle b_1 + a_1, b_2 + a_2 \rangle.$$

By the commutative property of addition:

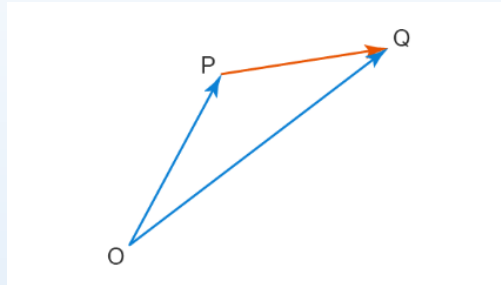
$$\vec{a} + \vec{b} = \vec{b} + \vec{a}.$$



Position Vectors

- A position vector describes the position of a point relative to an initial point or origin in a coordinate system.
- It specifies the displacement from the origin to the point of interest.
- The position vector of a point Q relative to P is given as:

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}.$$



Example

Find the vector with:

1. Initial point $A(5, 4)$ and terminal point $B(2, -1)$,
2. Initial point $B(3, -5)$ and terminal point $A(2, 7)$.



Solution

1. Given an initial point $A(5, 4)$ and a terminal point $B(2, -1)$:

$$\overrightarrow{AB} = B - A = \langle 2, -1 \rangle - \langle 5, 4 \rangle = \langle -3, -5 \rangle.$$

2. Given an initial point $B(3, -5)$ and a terminal point $A(2, 7)$:

$$\overrightarrow{BA} = A - B = \langle 2, 7 \rangle - \langle 3, -5 \rangle = \langle -1, 12 \rangle.$$



Unit Vector

Definition:

Any vector that has a unit length (magnitude) is called a unit vector.

Theorem:

For any non-zero vector $\vec{a} = \langle a_1, a_2 \rangle$, a unit vector having the same direction as $\vec{a}$ is given by:

$$\hat{u} = \frac{1}{\|\vec{a}\|} \vec{a}.$$

Notes:

1. The formula holds for any n -dimensional vector space.
2. The process of dividing a non-zero vector by its magnitude is called **normalization**.



Example

Find the unit vector in the same direction as $\vec{a} = \langle 4, -5 \rangle$.



Solution

Compute $\|\vec{a}\|$:

$$\|\vec{a}\| = \|\langle 4, -5 \rangle\| = \sqrt{4^2 + (-5)^2} = \sqrt{41}.$$

Compute the unit vector:

$$\hat{u} = \frac{1}{\|\vec{a}\|} \vec{a} = \frac{1}{\sqrt{41}} \langle 4, -5 \rangle.$$

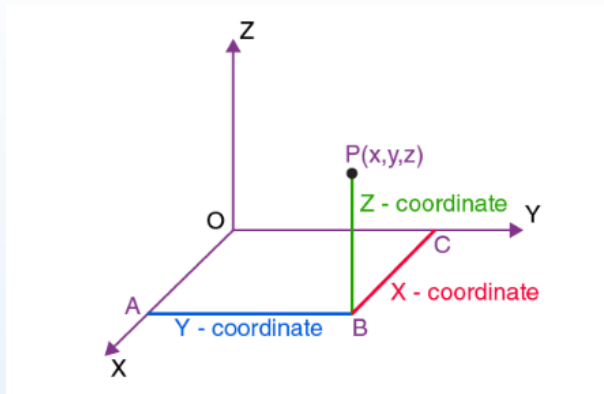
Therefore:

$$\hat{u} = \left\langle \frac{4}{\sqrt{41}}, \frac{-5}{\sqrt{41}} \right\rangle.$$



Vectors in Space (3D)

1. Consider the three dimension plane in the figure below



2. Given $\vec{a} = \langle a_1, a_2, a_3 \rangle$, $\vec{b} = \langle b_1, b_2, b_3 \rangle \in \mathbb{R}^3$,
3. The following properties hold:



- $\|\vec{a}\| = \|\langle a_1, a_2, a_3 \rangle\| = \sqrt{a_1^2 + a_2^2 + a_3^2}$
- $\vec{AB} = \langle b_1 - a_1, b_2 - a_2, b_3 - a_3 \rangle$
- $\vec{a} + \vec{b} = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$ (addition)
- $\vec{a} - \vec{b} = \langle a_1 - b_1, a_2 - b_2, a_3 - b_3 \rangle$ (subtraction)
- $c\vec{a} = \langle ca_1, ca_2, ca_3 \rangle$ (scalar multiplication).

Note: All other properties stated under 2-dimensional vector space hold.



Example

Find the unit vector in the same direction as $\vec{a} = \langle 1, -5, 2 \rangle$ and write $\vec{a}$ as the product of its magnitude and a unit vector.



Solution

Compute $\|\vec{a}\|$:

$$\|\vec{a}\| = \|\langle 1, -5, 2 \rangle\| = \sqrt{1^2 + (-5)^2 + 2^2} = \sqrt{30}.$$

Compute the unit vector:

$$\hat{u} = \frac{1}{\sqrt{30}} \langle 1, -5, 2 \rangle = \left\langle \frac{1}{\sqrt{30}}, \frac{-5}{\sqrt{30}}, \frac{2}{\sqrt{30}} \right\rangle.$$

Write $\vec{a}$ as the product of its magnitude and a unit vector:

$$\vec{a} = \left\langle \frac{1}{\sqrt{30}}, \frac{-5}{\sqrt{30}}, \frac{2}{\sqrt{30}} \right\rangle.$$



The Dot Product

Given the vectors $\vec{a} = \langle a_1, a_2, a_3 \rangle$, $\vec{b} = \langle b_1, b_2, b_3 \rangle \in \mathbb{R}^3$, the dot product is defined as:

$$\vec{a} \cdot \vec{b} = \langle a_1, a_2, a_3 \rangle \cdot \langle b_1, b_2, b_3 \rangle = a_1 b_1 + a_2 b_2 + a_3 b_3.$$

Similarly, for $\vec{a} = \langle a_1, a_2 \rangle$, $\vec{b} = \langle b_1, b_2 \rangle \in \mathbb{R}^2$:

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2.$$

Note: The dot product is also referred to as the scalar product.



Example

Compute the dot product $\vec{a} \cdot \vec{b}$ for:

- $\vec{a} = \langle 3, 2, -5 \rangle$ and $\vec{b} = \langle 3, -2, 4 \rangle$,
- $\vec{a} = \mathbf{i} - 3\mathbf{j}$ and $\vec{b} = 5\mathbf{i} + 2\mathbf{j}$.



Solution

For $\vec{a} = \langle 3, 2, -5 \rangle$ and $\vec{b} = \langle 3, -2, 4 \rangle$:

$$\vec{a} \cdot \vec{b} = 3(3) + 2(-2) + -5(4)$$

$$= 9 - 4 - 20$$

$$= -15$$

For $\vec{a} = \mathbf{i} - 3\mathbf{j}$ and $\vec{b} = 5\mathbf{i} + 2\mathbf{j}$:

$$\vec{a} \cdot \vec{b} = \langle 1, -3 \rangle \cdot \langle 5, 2 \rangle$$

$$= 1(5) + (-3)(2)$$

$$= 5 - 6$$

$$= -1$$



Trial Work

Given the three-dimensional basis vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$, find:

1. $\mathbf{i} \cdot \mathbf{i}$
2. $\mathbf{i} \cdot \mathbf{j}$
3. $\mathbf{i} \cdot \mathbf{k}$
4. $\mathbf{j} \cdot \mathbf{k}$
5. $\mathbf{j} \cdot \mathbf{j}$
6. $\mathbf{j} \cdot \mathbf{i}$
7. $\mathbf{k} \cdot \mathbf{j}$
8. $\mathbf{k} \cdot \mathbf{i}$
9. $\mathbf{k} \cdot \mathbf{k}$.



Properties of Dot Product

Consider the vectors $\vec{a}, \vec{b}, \vec{c} \in \mathbb{R}^n$ and any scalar $d \in \mathbb{R}$, the following hold:

- $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a},$
- $\vec{a} \cdot (\vec{b} + \vec{c}) = (\vec{a} \cdot \vec{b}) + (\vec{a} \cdot \vec{c}),$
- $(d\vec{a}) \cdot \vec{b} = d(\vec{a} \cdot \vec{b}) = \vec{a} \cdot (d\vec{b}),$
- $0 \cdot \vec{a} = 0,$
- $\vec{a} \cdot \vec{a} = \|\vec{a}\|^2.$



Angle Between Two Vectors

Theorem:

Let θ be the angle between the non-zero vectors $\vec{a}$ and $\vec{b}$. Then:

$$\vec{a} \cdot \vec{b} = \|\vec{a}\| \|\vec{b}\| \cos \theta.$$

Thus:

$$\theta = \cos^{-1} \left(\frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \|\vec{b}\|} \right).$$



Example

Find the angle between the vectors:

$$\vec{a} = \langle 3, -2, 4 \rangle \quad \text{and} \quad \vec{b} = \langle 1, 2, 5 \rangle.$$



Solution

Using the formula:

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \|\vec{b}\|},$$

compute the dot product:

$$\vec{a} \cdot \vec{b} = 3(1) + (-2)(2) + 4(5) = 19.$$

Compute the magnitudes:

$$\|\vec{a}\| = \sqrt{3^2 + (-2)^2 + 4^2} = \sqrt{29}, \quad \|\vec{b}\| = \sqrt{1^2 + 2^2 + 5^2} = \sqrt{30}.$$

Substituting:

$$\cos \theta = \frac{19}{\sqrt{29} \cdot \sqrt{30}} = \frac{19}{\sqrt{870}}.$$

Thus:

$$\theta = \cos^{-1} \left(\frac{19}{\sqrt{870}} \right).$$



Orthogonal Vectors

Corollary

Two non-zero vectors $\vec{a}$ and $\vec{b}$ are said to be orthogonal if and only if:

$$\vec{a} \cdot \vec{b} = 0.$$



Example

Determine whether the following pairs of vectors are orthogonal:

1. $\vec{p} = \langle 1, 3, -5 \rangle$ and $\vec{q} = \langle 3, -1, 0 \rangle$

2. $\vec{m} = \langle 4, 4, 8 \rangle$ and $\vec{n} = \langle 3, 12, 2 \rangle$.



Solution

1. Compute the dot product of $\vec{p}$ and $\vec{q}$:

$$\begin{aligned}\vec{p} \cdot \vec{q} &= 3 - 3 + 0 \\ &= 0.\end{aligned}$$

Therefore, the vectors $\vec{p}$ and $\vec{q}$ are **orthogonal**.

1. Compute the dot product of $\vec{m}$ and $\vec{n}$:

$$\begin{aligned}\vec{m} \cdot \vec{n} &= 12 + 48 + 16 \\ &= 76 \neq 0\end{aligned}$$

Therefore, the vectors $\vec{m}$ and $\vec{n}$ are **not orthogonal**.



Orthonormal Vectors

- A set of vectors is said to be an orthonormal set if every vector is a **unit vector** and every vector is **orthogonal** to the others.
- In a three-dimensional space, the vectors:

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

are **orthonormal** because they are **orthogonal** to each other and each has a magnitude of 1.



Example

Verify if the following set of vectors is an orthonormal set:

$$\mathbf{u}_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}, \quad \text{and} \quad \mathbf{u}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$



Solution

1. Checking for Orthogonality:

-

$$\mathbf{u}_1 \cdot \mathbf{u}_2 = \left(\frac{1}{\sqrt{2}}\right) \left(\frac{1}{\sqrt{2}}\right) + \left(\frac{1}{\sqrt{2}}\right) \left(-\frac{1}{\sqrt{2}}\right) + 0 \cdot 0 = \frac{1}{2} - \frac{1}{2} = 0$$

- $\mathbf{u}_1 \cdot \mathbf{u}_3 = 0$

- $\mathbf{u}_2 \cdot \mathbf{u}_3 = 0$



2. Verifying the vectors are Unit Vectors

- $\mathbf{u}_1 \cdot \mathbf{u}_1 = \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + 0 = \frac{1}{2} + \frac{1}{2} = 1$
- $\mathbf{u}_2 \cdot \mathbf{u}_2 = \left(\frac{1}{\sqrt{2}}\right)^2 + \left(-\frac{1}{\sqrt{2}}\right)^2 + 0 = \frac{1}{2} + \frac{1}{2} = 1$
- $\mathbf{u}_3 \cdot \mathbf{u}_3 = 0^2 + 0^2 + 1^2 = 1.$

Since both conditions are met, the vectors form an orthonormal set.



Continue from Lecture 1B

